

PVsyst - Simulation report

Grid-Connected System

Project: caso1-bom-jesus

Variant: New simulation variant

No 3D scene defined, no shadings

System power: 2400 Wp

Bom Jesus da Lapa - Brazil

**PVsyst V8.1.5**

VCO, Simulation date:
09/09/26 16:32
with V8.1.5

Project summary**Geographical Site**

Bom Jesus da Lapa
Brazil

Situation

Latitude -13.2499 °(S)
Longitude -43.4101 °(W)
Altitude 443 m
Time zone UTC-3

Project settings

Albedo 0.20

Weather data

Bom Jesus da Lapa
NASA POWER satellite data 2001-2020 - Synthetic

System summary**Grid-Connected System****Orientation #1**
Horizontal plane**System information****PV Array**

Nb. of modules
Pnom total

No 3D scene defined, no shadings**Near Shadings**

no Shadings

Inverters

Nb. of units
Total power
Pnom ratio

User's needs

Unlimited load (grid)

4 units
2400 Wp

1 unit
2 kWac
1.20

Results summary

Simulation step Hourly

Produced Energy 4411.8 kWh/year Specific production 1838 kWh/kWp/year Perf. Ratio PR 83.35 %

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General parameters**Grid-Connected System**

No 3D scene defined, no shadings

Simulation step

Hourly

Orientation #1

Horizontal plane

Models used

Transposition	Perez
Diffuse	Perez, Meteonorm
Circumsolar	separate

Horizon

Free Horizon

Near Shadings

no Shadings

User's needs

Unlimited load (grid)

PV Array Characteristics**PV module**

Manufacturer	Generic
Model	JKM-600N-72HL4-V
(Original PVsyst database)	

Unit Nom. Power	600 Wp
Number of PV modules	4 units
Nominal (STC)	2400 Wp
Modules	1 unit x 4 In series

At operating cond. (50°C)

Pmpp	2226 Wp
U mpp	163 V
I mpp	14 A

Total PV power

Nominal (STC)	2.40 kWp
Total	4 modules
Module area	10.3 m ²
Cell area	9.5 m ²

Inverter

Manufacturer	Generic
Model	MIC 2000TL-X
(Original PVsyst database)	

Unit Nom. Power	2.00 kWac
Number of inverters	1 unit
Total power	2.0 kWac
Operating voltage	50-500 V
Pnom ratio (DC:AC)	1.20

Total inverter power

Total power	2 kWac
Number of inverters	1 unit
Pnom ratio	1.20

Array losses**Thermal Loss factor**

Module temperature according to irradiance	
Uc (const)	20.0 W/m ² K
Uv (wind)	0.0 W/m ² K/m/s

DC wiring losses

Global array res.	196 mΩ
Loss Fraction	1.50 % at STC

Module Quality Loss

Loss Fraction	-0.75 %
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Module mismatch losses

Loss Fraction	2.00 % at MPP
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IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.963	0.892	0.814	0.679	0.438	0.000



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Main results

Simulation step

Hourly

System Production

Produced Energy

4411.8 kWh/year

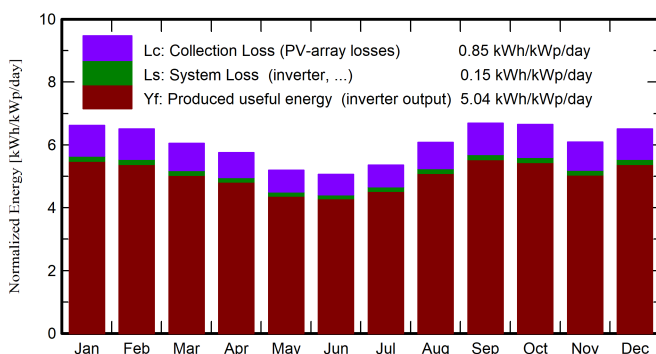
Specific production

1838 kWh/kWp/year

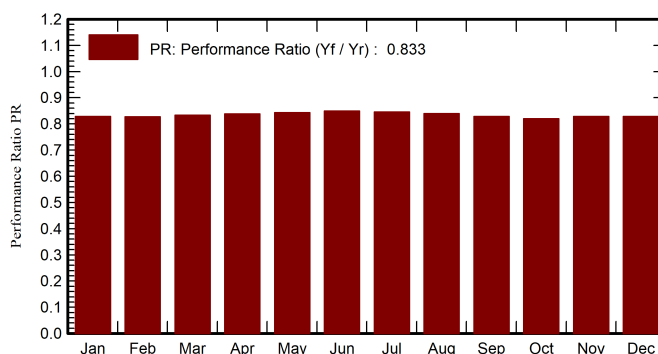
Perf. Ratio PR

83.35 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor	DiffHor	T_Amb	GlobInc	GlobEff	EArray	E_Grid	PR
	kWh/m ²	kWh/m ²	°C	kWh/m ²	kWh/m ²	kWh	kWh	ratio
January	205.4	74.17	27.13	205.3	200.6	420.5	408.1	0.828
February	182.2	63.25	27.14	182.1	178.0	372.7	361.7	0.828
March	187.5	65.82	27.23	187.5	182.9	386.4	375.1	0.834
April	172.6	52.52	26.86	172.6	168.5	357.7	347.2	0.838
May	160.9	43.62	26.42	160.9	156.3	335.5	325.6	0.843
June	151.7	34.62	25.12	151.7	147.0	318.5	309.1	0.849
July	166.2	34.72	24.57	166.1	161.2	347.4	337.1	0.846
August	188.4	39.60	25.67	188.4	183.9	391.2	379.8	0.840
September	200.6	46.39	27.71	200.6	196.2	410.5	398.6	0.828
October	206.1	59.74	29.03	206.0	201.6	417.8	405.4	0.820
November	182.8	68.36	27.60	182.8	178.6	374.5	363.4	0.828
December	201.7	71.12	27.14	201.6	197.0	412.8	400.7	0.828
Year	2206.1	653.93	26.80	2205.6	2151.9	4545.6	4411.8	0.833

Legends

GlobHor Global horizontal irradiation

DiffHor Horizontal diffuse irradiation

T_Amb Ambient Temperature

GlobInc Global incident in coll. plane

GlobEff Effective Global, corr. for IAM and shadings

EArray Effective energy at the output of the array

E_Grid Energy injected into grid

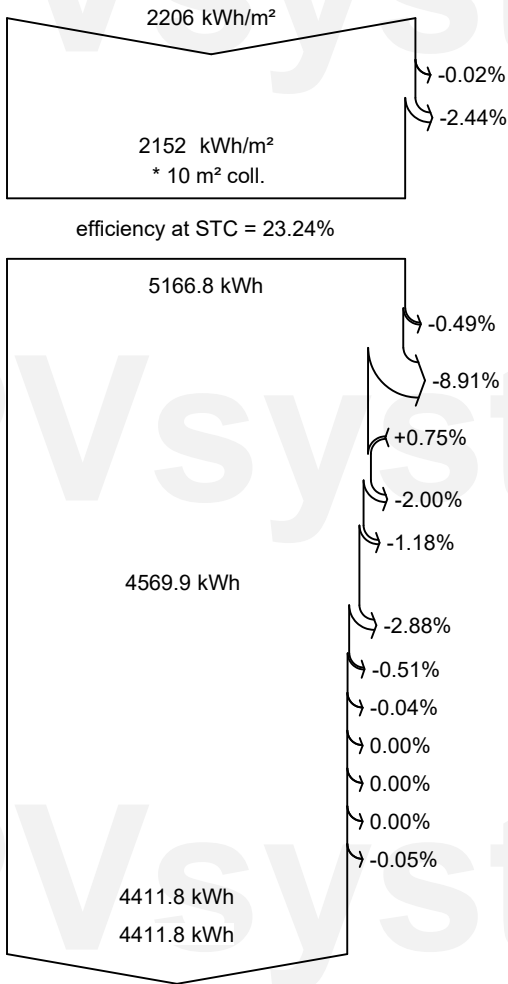
PR Performance Ratio



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Loss diagram



Global horizontal irradiation

Global incident in coll. plane

IAM factor on global

Effective irradiation on collectors

PV conversion

Array nominal energy (at STC effic.)

PV loss due to irradiance level

PV loss due to temperature

Module quality loss

Module array mismatch loss

Ohmic wiring loss

Array virtual energy at MPP

Inverter Loss during operation (efficiency)

Inverter Loss over nominal inv. power

Inverter Loss due to max. input current

Inverter Loss over nominal inv. voltage

Inverter Loss due to power threshold

Inverter Loss due to voltage threshold

Night consumption

Available Energy at Inverter Output

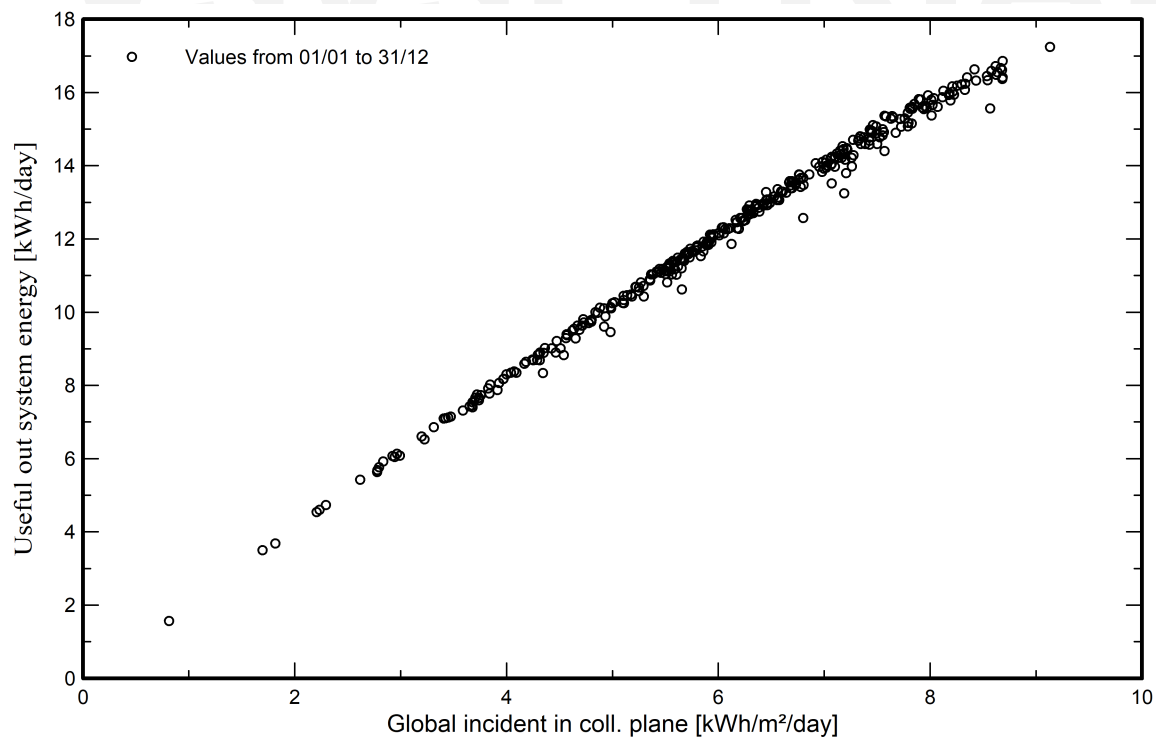
Energy injected into grid



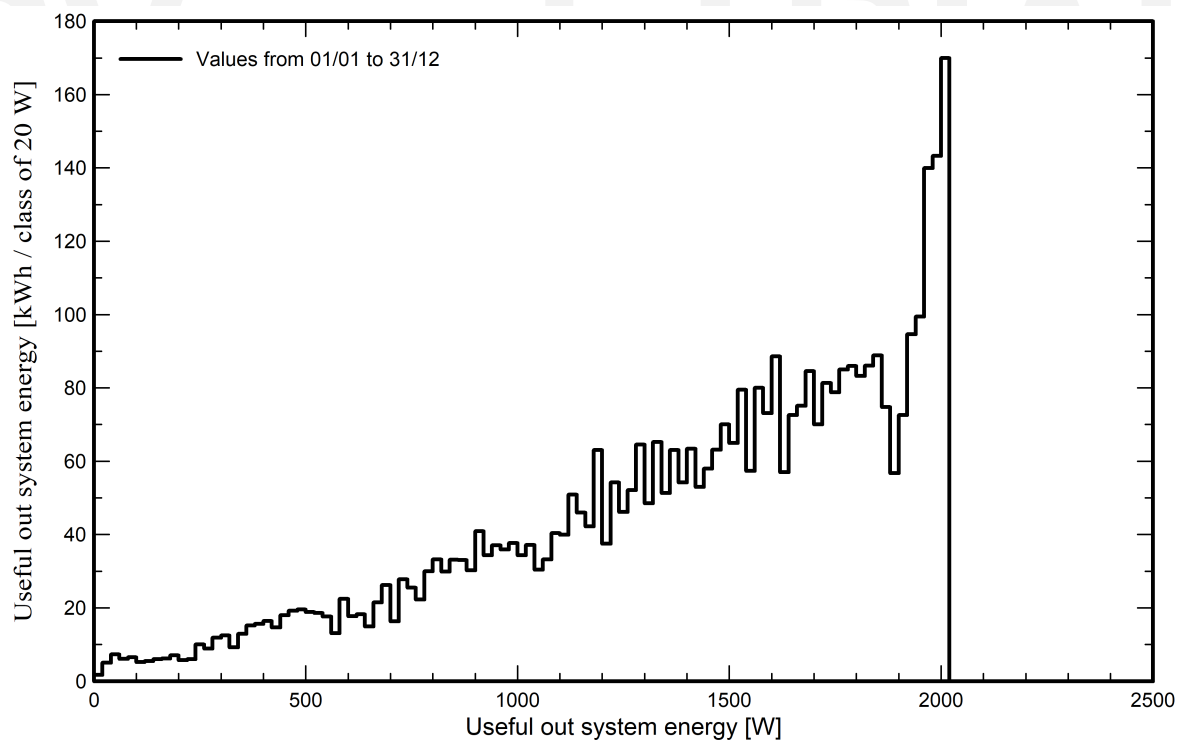
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Predef. graphs
Daily Input/Output diagram



System Output Power Distribution

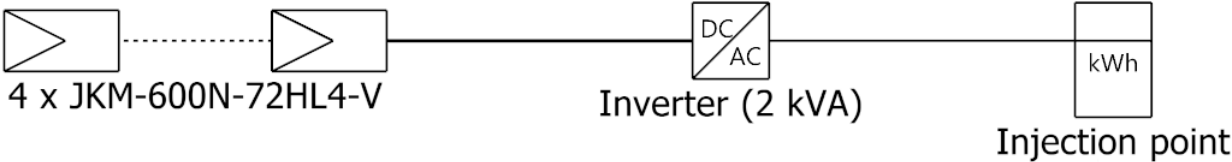




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Single-line diagram



PV module	JKM-600N-72HL4-V
Inverter	MIC 2000TL-X
String	4 x JKM-600N-72HL4-V

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VC0 : New simulation variant

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